AP PRECALCULUS

Unit 1 • Lesson 1.1 practice • Answer key

Lesson 1.1 • Change in Tandem

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The table gives values of f. On which interval does the table show f decreasing?

x	-2	-1	0	1	2
$f(x)$	7	2	-1	-2	-1

(A) $[-2, 1]$
(B) $[-1, 2]$
(C) $[0, 2]$
(D) $[-2, 2]$

Solution

Answer: (A). The outputs decrease from 7 to -2 as x goes from -2 to 1, then increase.

Question 2

NO CALCULATOR

What are the zeros of $g(x) = (x - 4)(x + 1)$?

(A) -4 and 1
(B) 4 only
(C) -1 and 4
(D) 0 and 4

Solution

Answer: (C). $g(x) = 0$ when $x - 4 = 0$ or $x + 1 = 0$.

Question 3

NO CALCULATOR

A function is increasing and its rate of change is decreasing. Which describes its graph?

(A) Increasing and concave up
(B) Increasing and concave down
(C) Decreasing and concave up
(D) Decreasing and concave down

Solution

Answer: (B). A decreasing rate of change means the graph is concave down.

Question 4

NO CALCULATOR

A car's distance d (meters) from a stop sign is recorded: $d(0) = 0$, $d(1) = 10$, $d(2) = 40$, $d(3) = 90$. Which statement fits the data?

(A) d is decreasing and concave up.

(B) d is increasing and concave down.

(C) d is linear.

(D) d is increasing and concave up.

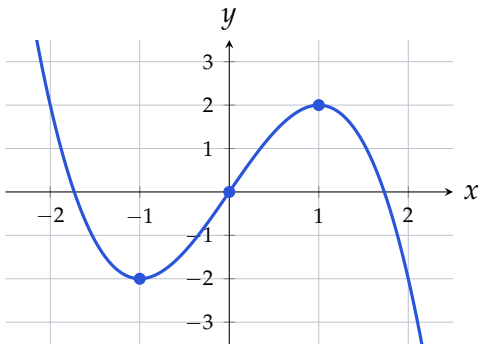
Solution

Answer: (D). The changes $10, 30, 50$ are positive and increasing: increasing and concave up.

Question 5 (graph)

NO CALCULATOR

The graph of $f(x) = -x^3 + 3x$ is shown, with turning points at $(-1, -2)$ and $(1, 2)$. On which interval is f both increasing and concave down?



(A) $(-1, 0)$

(B) $(0, 1)$

(C) $(1, 2)$

(D) $(-2, -1)$

Solution

Answer: (B). f is increasing between its turning points, on $(-1, 1)$. The graph bends downward (concave down) to the right of the inflection point $(0, 0)$. Both hold on $(0, 1)$.

Question 6 (Free response)

NO CALCULATOR

The height h, in meters, of a ball is recorded every half second.

t (s)	0	0.5	1	1.5	2
$h(t)$ (m)	2	9	12	11	6

(a) Find the change in h over each half-second interval.

(b) Based on these changes, is the graph of h concave up or concave down on $[0, 2]$? Give a reason.

(c) Between which two recorded times does h change from increasing to decreasing, according to the table?

Solution

(a) $9 - 2 = 7$, $12 - 9 = 3$, $11 - 12 = -1$, $6 - 11 = -5$.

(b) The changes over equal intervals are decreasing ($7 > 3 > -1 > -5$), so the rate of change is decreasing: concave down.

(c) The changes go from positive ($+3$ on $[0.5, 1]$) to negative (-1 on $[1, 1.5]$), so the ball reaches

its maximum height between $t = 0.5$ and $t = 1.5$; the table shows h increasing up to $t = 1$ and decreasing after it.